

§11.5

Vector valued function $F: \mathbb{R} \rightarrow \mathbb{R}^3$

$$\vec{F}(t) = (f(t), g(t), h(t)) = \vec{i} f(t) + \vec{j} g(t) + \vec{k} h(t)$$

eg $\vec{F}(t) = (t, \cos t, \sin t)$

Defn $\lim_{t \rightarrow c} \vec{F}(t) = \vec{L}$ means: for any $\varepsilon > 0$
there is a $\delta > 0$ such that $\|\vec{F}(t) - \vec{L}\| < \varepsilon$
provided $0 < |t - c| < \delta$.

In other words $0 < |t - c| < \delta \Rightarrow \|\vec{F}(t) - \vec{L}\| < \varepsilon$

Thm $\vec{F}(t)$ has a limit if and only if
 $f(t), g(t), h(t)$ have limits.

Proof $\Leftarrow$ Assume $0 < |t - c| < \delta_1 \Rightarrow |f(t) - L_1| < \frac{\varepsilon}{\sqrt{3}}, 0 < |t - c| < \delta_2$

$$\Rightarrow |g(t) - L_2| < \frac{\varepsilon}{\sqrt{3}} \text{ and } 0 < |t - c| < \delta_3 \Rightarrow |h(t) - L_3| < \frac{\varepsilon}{\sqrt{3}}$$

Let $\delta = \min \{ \delta_1, \delta_2, \delta_3 \}$, then

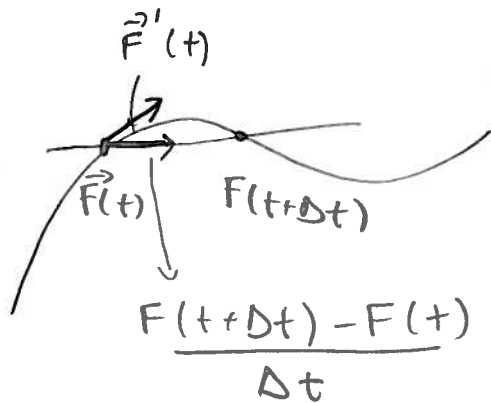
$$\|\vec{F}(t) - \vec{L}\| = ((f(t) - L_1)^2 + (g(t) - L_2)^2 + (h(t) - L_3)^2)^{1/2} \leq \left(3 \frac{\varepsilon^2}{3} \right)^{1/2} \leq \varepsilon$$

$\Rightarrow$ Assume $0 < |t - c| < \delta \Rightarrow \|\vec{F}(t) - \vec{L}\| \leq \varepsilon$, then

$$|f(t) - L_1| \leq ((f(t) - L_1)^2 + (g(t) - L_2)^2 + (h(t) - L_3)^2)^{1/2} = \|\vec{F}(t) - \vec{L}\| \leq \varepsilon$$

(and similarly for $|g(t) - L_2|$ & $|h(t) - L_3|$)

Defn $\vec{F}'(t) = (f'(t), g'(t), h'(t)) = \lim_{\Delta t \rightarrow 0} \frac{\vec{F}(t+\Delta t) - \vec{F}(t)}{\Delta t}$



$$D_t \vec{F}(t) = \vec{F}'(t)$$

NOTATION.

Eg $(\cos t, t^2, e^t)' = (-\sin t, 2t, e^t)$

Rules ① $[\vec{F}(t) + \vec{G}(t)]' = \vec{F}'(t) + \vec{G}'(t)$

② $[c \cdot \vec{F}(t)]' = c \vec{F}'(t)$

③ $D_t [p(t) \vec{F}(t)] = p(t) \cdot \vec{F}'(t) + p'(t) \vec{F}(t)$

④ $D_t [\vec{F}(t) \cdot \vec{G}(t)] = \vec{F}'(t) \cdot \vec{G}(t) + \vec{F}(t) \cdot \vec{G}'(t)$

⑤ $D_t [\vec{F}(t) \times \vec{G}(t)] = \vec{F}(t) \times \vec{G}'(t) + \vec{F}'(t) \times \vec{G}(t)$

⑥ $D_t [\vec{F}(g(t))] = \vec{F}'(g(t)) \cdot p'(t)$

Eg $\vec{F}(t) = (\cos t, \sin t, t)$ $D_t [\vec{F}(t) \cdot \vec{F}'(t)] =$

$= D_t [(\cos t, \sin t, t) \cdot (-\sin t, \cos t, 1)] = D_t (1) = 1$

$\vec{F}'(t) \cdot \vec{F}'(t) + \vec{F}'(t) \cdot \vec{F}''(t) = (-\sin t, \cos t, 1) \cdot (-\sin t, \cos t, 1) + (-\sin t, \cos t, 1) \cdot (-\cos t, -\sin t, 0)$

$= -\sin^2 t + \cos^2 t + 1 - \cos^2 t - \sin^2 t + 0 = 1$

$$\int_0^1 (t, t^2, te^{t^2}) dt = \left(\left. \frac{t^2}{2} \right|_0^1, \left. \frac{t^3}{3} \right|_0^1, \left. e^{t^2} \right|_0^1 \right) = \left(\frac{1}{2}, \frac{1}{3}, e-1 \right)$$

$$D_t [t \vec{F}(t)] = \vec{F}(t) + t \vec{F}'(t)$$

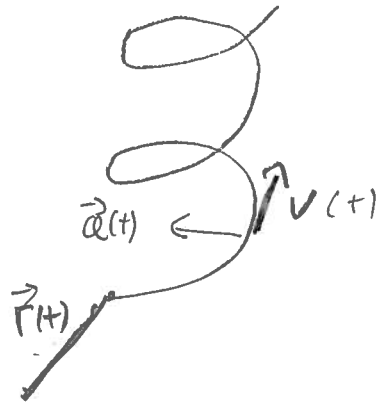
$\vec{r}(t) = (f(t), g(t), h(t))$ the position vector

$\vec{v}(t) = \vec{r}'(t) = (f'(t), g'(t), h'(t))$ the velocity vector

$\vec{a}(t) = \vec{r}''(t) = (f''(t), g''(t), h''(t))$ the acceleration vector

Ex $\vec{r}(t) = (\cos t, \sin t, t) \quad \vec{v}(t) = (-\sin t, \cos t, 1)$

$\vec{a}(t) = (-\cos t, -\sin t, 0)$



$$s = \text{speed} = \sqrt{(f'(t))^2 + (g'(t))^2 + (h'(t))^2} = \|\vec{v}(t)\|$$

$$L = \int_a^b s(u) du$$

$$L(t) = \int_a^t s(u) du$$

$$\frac{dL(t)}{dt} = s(t)$$

Fundamental theorem
of calculus

Ex $\|\vec{v}(t)\| = \text{constant} = C$

Then $\vec{v}(t) \cdot \vec{v}(t) = C^2$ so

$$0 = D_t(\vec{v}(t) \cdot \vec{v}(t)) = \vec{v}'(t) \cdot \vec{v}(t) + \vec{v}(t) \cdot \vec{v}'(t) = 2\vec{v}(t) \cdot \vec{a}(t)$$

If the speed is constant, then $\vec{a}$ and $\vec{v}$ are perpendicular.

Ex $\vec{r}(t) = (a \cos \omega t, a \sin \omega t)$

$$\vec{v}(t) = (-a\omega \sin \omega t, a\omega \cos \omega t)$$

$$\vec{a}(t) = (-a\omega^2 \cos \omega t, -a\omega^2 \sin \omega t)$$

$$s(t) = (a^2 \omega^2 \sin^2 \omega t + a^2 \omega^2 \cos^2 \omega t)^{1/2} = (a^2 \omega^2)^{1/2} = a\omega$$

$$\vec{v}(t) \cdot \vec{a}(t) = +a^2 \omega^3 \sin \omega t \cos \omega t - a^2 \omega^3 \cos \omega t \sin \omega t = 0$$

Ex $\frac{x^2}{4} + \frac{y^2}{9} = 1$ is an ellipse.

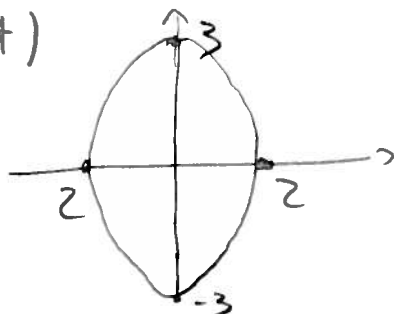
$\vec{r}(t) = (2 \cos t, 3 \sin t)$ gives the ellipse in parametric form

$$\vec{v}(t) = (-2 \sin t, 3 \cos t)$$

$$s(t) = (4 \sin^2 t + 9 \cos^2 t)^{1/2} = (4 + 5 \cos^2 t)^{1/2}$$

$$\vec{a}(t) = (-2 \cos t, -3 \sin t)$$

$\nearrow 4 = \min$
for $t = \pi/2, 3\pi/2$
 $\searrow 9 = \max$ for $t = 0, \pi$

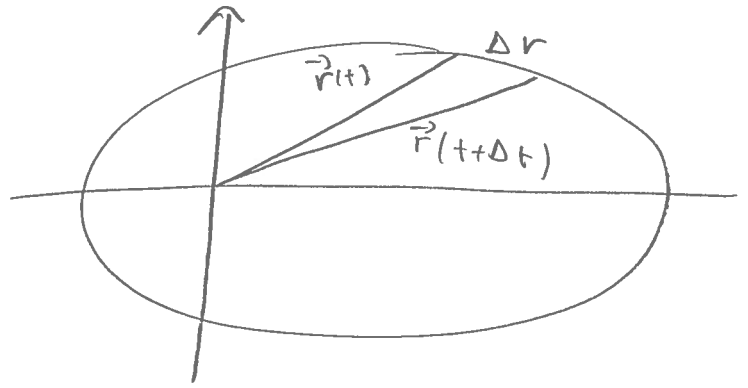


Kepler's 2nd LAW

$$\Delta A \approx \frac{1}{2} \|\vec{r}(t) \times \Delta \vec{r}\|$$

$$\frac{\Delta A}{\Delta t} \approx \frac{1}{2} \|\vec{r}(t) \times \frac{\Delta \vec{r}}{\Delta t}\|$$

$$\frac{dA}{dt} = \frac{1}{2} \|\vec{r}(t) \times \vec{v}(t)\|$$



Now $\frac{d}{dt} (\vec{r}(t) \times \vec{v}(t)) = \underbrace{\vec{v}(t) \times \vec{v}(t)}_0 + \underbrace{\vec{r}(t) \times \vec{a}(t)}_0 \text{ as } \vec{a}(t) = -\frac{GM}{\|\vec{r}(t)\|^3} \vec{r}(t)$

So $\vec{r}(t) \times \vec{v}(t)$ is constant

Ex Find the limit if it exists:

a) $\lim_{t \rightarrow 1} \left[\frac{t-1}{t^2-1}, -\left(\frac{t^2+2t-3}{t-1} \right) \right]$ b) $\lim_{t \rightarrow 0} \left(\frac{\sin t \cos t}{t}, \frac{-7t^3}{e^t}, \frac{t}{t+1} \right)$

c) $\lim_{t \rightarrow 0^+} \vec{i} \ln(t^3) + \vec{j} t^2 \ln t + \vec{k} t$

Ex Find the domain of a) $\vec{r}(t) = \left(\frac{2}{t-4}, \sqrt{3-t}, \ln|4-t| \right)$
b) $\vec{r}(t) = \cos t \vec{i} + \sin t \vec{j} + \sqrt{9-t^2} \vec{k}$

Ex If $\vec{r}(t) = (e^{-t}, -\ln(t^2))$ find $D_t [\vec{r}(t) \cdot \vec{r}''(t)]$

Ex Find velocity $\vec{v}$ and acceleration $\vec{a}$ and speeds

for a) $\vec{r}(t) = 4t \vec{i} + 5(t^2-1) \vec{j} + 2t \vec{k}$

b) $\vec{r}(t) = (1/t, (t^2-1)^{-1}, t^5)$

c) $\vec{r}(t) = (\tan(t), 3e^t, \cos 4t)$

Ex Find the length of the curve

a) $\vec{r}(t) = (t \cos t, t \sin t, \sqrt{2}t)$ $0 \leq t \leq 2$

b) $\vec{r}(t) = (\sqrt{6}t^2, \frac{2}{3}t^3, 6t)$ $3 \leq t \leq 6$